

Architectural design

Multiple Choice Questions

1. What does architectural design primarily define in software engineering?
 - A) Programming language
 - B) System architecture and structure
 - C) User experience
 - D) Deployment scripts
2. Which of the following is **not** a common layer in layered architecture?
 - A) Presentation
 - B) Transport
 - C) Data Access
 - D) Business Logic
3. The pipe-and-filter architecture is best suited for:
 - A) Image rendering
 - B) Incremental data processing
 - C) Relational database design
 - D) UI layout design
4. In pull-based data flow, the process starts from the:
 - A) Source
 - B) Buffer
 - C) Sink
 - D) UI
5. Which component handles business rules and logic?
 - A) Data Access
 - B) User Interface
 - C) Business Logic
 - D) Controller
6. What type of connection do filters typically have in a pipe-and-filter model?
 - A) Multi-threaded queues
 - B) Sockets

- C) Pipes
 - D) Events
7. In a layered system, which layer is responsible for data presentation?
- A) Controller
 - B) Middleware
 - C) Presentation
 - D) Logic
8. The main benefit of layered architecture is:
- A) Obfuscation
 - B) Hard coding
 - C) Modularity and separation of concerns
 - D) Minimizing files
9. What function is typically called to move data forward in a push model?
- A) Pull()
 - B) Process()
 - C) Execute()
 - D) Write()
10. The model-view-controller (MVC) pattern does NOT include which component?
- A) Model
 - B) View
 - C) Architecture
 - D) Controller
11. Which of the following acts as a data consumer in a pipeline?
- A) Filter
 - B) Sink
 - C) Source
 - D) Buffer
12. In a pull model, each component typically calls the:
- A) source.Read()
 - B) sink.Read()
 - C) filter.Write()
 - D) model.Pull()
13. What is the purpose of buffering in data pipes?
- A) To increase latency
 - B) To synchronize data flow
 - C) To create concurrency
 - D) To ensure UI control

14. Which UI component appears in the architecture?
- A) UserForm
 - B) Order UI
 - C) ViewController
 - D) PipeUI
15. What is the correct order of data flow in pipe-and-filter?
- A) Source $\rightarrow$ Filter $\rightarrow$ Sink
 - B) Filter $\rightarrow$ Source $\rightarrow$ Sink
 - C) Sink $\rightarrow$ Filter $\rightarrow$ Source
 - D) UI $\rightarrow$ Logic $\rightarrow$ Access
16. Which type of chart is used to display numerical breakdowns in the presentation?
- A) Radar chart
 - B) Line chart
 - C) Pie chart
 - D) Flowchart
17. Which class is included in the sample architecture diagram?
- A) ClassY
 - B) ClassH
 - C) ClassN
 - D) ClassB
18. A good architectural design promotes:
- A) Tight coupling
 - B) Performance tradeoffs
 - C) Maintainability and scalability
 - D) Hardcoded paths
19. Which layer stores and retrieves data?
- A) Presentation
 - B) Controller
 - C) Data Access
 - D) Middleware
20. In the MVC pattern, which part manages data and logic?
- A) View
 - B) Filter
 - C) Model
 - D) Presentation
21. What typically initiates data flow in push-based architecture?
- A) Sink

- B) UI
 - C) Source
 - D) Controller
22. Which component belongs to business logic in the architecture?
- A) ShoppingBasket B. Logic
 - B) ShoppingBasket UI
 - C) ShoppingBasket View
 - D) ShoppingBasket Model
23. Filters in pipe-and-filter architecture should be:
- A) Stateful and persistent
 - B) Tightly coupled
 - C) Independent and reusable
 - D) Event-driven
24. What does "UI" stand for in the context of layered architecture?
- A) Unified Infrastructure
 - B) User Interface
 - C) Universal Integration
 - D) Unstructured Input
25. The function Run () in a pipeline typically:
- A) Formats UI elements
 - B) Loops through input data and writes output
 - C) Monitors network activity
 - D) Queries SQL databases
26. Which component is a data producer in a data flow system?
- A) Buffer
 - B) Sink
 - C) Source
 - D) Controller
27. Which of the following are **examples of functional domains** in the system?
- A) Order
 - B) Report
 - C) Notification
 - D) Payment
28. Which layer is closest to the end-user?
- A) Data Access
 - B) Presentation
 - C) Logic

D) Source

29. What is the role of the `Write(data)` function in the pipeline?

A) Displays UI elements

B) Sends data to the next filter

C) Loads configuration

D) Maps commands

30. A key requirement for future system evolution is:

A) Immutable classes

B) Static functions

C) Flexible architecture

D) Rigid structure

Answer Key

1. B	2. B	3. B	4. C	5. C
6. C	7. C	8. C	9. D	10. C
11. B	12. A	13. B	14. B	15. A
16. C	17. A	18. C	19. C	20. C
21. C	22. A	23. C	24. B	25. B
26. C	27. A	28. B	29. B	30. C

Multiple-Answer Questions (Select All That Apply)

1. Which of the following are architectural styles?
 - ☐ Pipe-and-filter
 - ☐ Layered
 - ☐ MVC
 - ☐ Pseudo-random
2. The business logic layer typically:
 - ☐ Handles data input/output
 - ☐ Applies domain rules
 - ☐ Validates business processes
 - ☐ Manages screen rendering
3. In a layered architecture, which layers are commonly used?
 - ☐ Presentation
 - ☐ Hardware
 - ☐ Business Logic
 - ☐ Data Access
4. Which components are used in the data flow model?
 - ☐ Source
 - ☐ Filter
 - ☐ Sink
 - ☐ User
5. Characteristics of the pipe-and-filter architecture include:
 - ☐ Independent filters
 - ☐ Shared memory
 - ☐ Reusable components
 - ☐ Parallel processing
6. The push-based model:
 - ☐ Requires the sink to request data
 - ☐ Pushes data automatically
 - ☐ Is initiated by the source
 - ☐ Operates on demand
7. The pull-based model:
 - ☐ Requires a scheduler
 - ☐ Operates from sink to source
 - ☐ Processes data reactively
 - ☐ Pushes data immediately
8. Which are examples of user interface components in the system?

- ☐ Order UI
- ☐ Customer UI
- ☐ Invoicing UI
- ☐ Filter1

9. MVC pattern consists of:

- ☐ Model
- ☐ View
- ☐ Code
- ☐ Controller

10. The presentation layer is responsible for:

- ☐ Handling user input
- ☐ Data validation
- ☐ Formatting output
- ☐ Managing business logic

11. Architectural design affects:

- ☐ Maintainability
- ☐ Scalability
- ☐ Typing speed
- ☐ Evolution

12. What are typical data flow directions in architecture models?

- ☐ Push
- ☐ Pull
- ☐ Sync
- ☐ Both push and pull

13. Business logic can reside in:

- ☐ Business Logic Layer
- ☐ UI Layer
- ☐ Data Layer
- ☐ Middleware

14. Examples of system domains shown in the document:

- ☐ Payment
- ☐ ShoppingBasket
- ☐ Scheduler
- ☐ Order

15. Which are valid data representations in the diagrams?

- ☐ Pie chart
- ☐ Column chart

- ☐ Flowchart
- ☐ Table chart

16. Filters in a pipeline:

- ☐ May be stateless
- ☐ Are tightly coupled
- ☐ Process data
- ☐ Use control flow

17. Architectural design specifies:

- ☐ Structure
- ☐ Behavior
- ☐ Implementation details
- ☐ Component interaction

18. Benefits of layered architecture include:

- ☐ Better organization
- ☐ Tighter integration
- ☐ Enhanced maintainability
- ☐ Loose coupling

19. Filters in a push architecture:

- ☐ Wait for data requests
- ☐ Push data to next stage
- ☐ Are initiated by upstream
- ☐ Depend on sink

20. Which classes were included in the architecture diagram?

- ☐ ClassX
- ☐ ClassY
- ☐ ClassC
- ☐ ClassZ

21. Which of the following can be considered data sinks?

- ☐ UI
- ☐ Final processing unit
- ☐ Output storage
- ☐ Source filters

22. In the diagrams, data flow is shown between:

- ☐ Source → Filter1 → Filter2 → Sink
- ☐ Filter1 → Source → Filter2
- ☐ UI → Business Logic → Data Access
- ☐ Business Logic → UI

23. System evolution requires:
- ☐ Flexible architecture
 - ☐ Monolithic design
 - ☐ Modular design
 - ☐ Layer independence
24. Which elements were labeled as part of business logic?
- ☐ Invoicing B. Logic
 - ☐ ShoppingBasket B. Logic
 - ☐ Order B. Logic
 - ☐ Customer View
25. Buffering pipes are used to:
- ☐ Speed up networks
 - ☐ Handle timing issues
 - ☐ Separate filters
 - ☐ Reduce memory usage
26. Data in pipe-and-filter systems can be:
- ☐ Transformed
 - ☐ Stored
 - ☐ Compressed
 - ☐ Filtered
27. Which data flows are shown with "Write(data)" functions?
- ☐ Push filters
 - ☐ Source to Filter1
 - ☐ Sink to Source
 - ☐ Filter2 to Sink
28. Which modules are part of data access in the diagrams?
- ☐ Order Data Access
 - ☐ Payment Data Access
 - ☐ ShoppingBasket Data Access
 - ☐ Thread Management
29. A sink typically:
- ☐ Collects data
 - ☐ Sends data
 - ☐ Discards data
 - ☐ Stores final results
30. A good software architecture should:
- ☐ Be maintainable
 - ☐ Reduce redundancy
 - ☐ Allow easy updates
 - ☐ Be hardcoded

Answer Key

1. ☒ Pipe-and-filter, ☒ Layered, ☒ MVC
2. ☒ Applies domain rules, ☒ Validates business processes
3. ☒ Presentation, ☒ Business Logic, ☒ Data Access
4. ☒ Source, ☒ Filter, ☒ Sink
5. ☒ Independent filters, ☒ Reusable components, ☒ Parallel processing
6. ☒ Pushes data automatically, ☒ Is initiated by the source
7. ☒ Operates from sink to source, ☒ Processes data reactively
8. ☒ Order UI, ☒ Customer UI, ☒ Invoicing UI
9. ☒ Model, ☒ View, ☒ Controller
10. ☒ Handling user input, ☒ Formatting output
11. ☒ Maintainability, ☒ Scalability, ☒ Evolution
12. ☒ Push, ☒ Pull, ☒ Both push and pull
13. ☒ Business Logic Layer, ☒ Middleware
14. ☒ Payment, ☒ ShoppingBasket, ☒ Order
15. ☒ Pie chart, ☒ Column chart, ☒ Table chart
16. ☒ May be stateless, ☒ Process data
17. ☒ Structure, ☒ Behavior, ☒ Component interaction
18. ☒ Better organization, ☒ Enhanced maintainability, ☒ Loose coupling
19. ☒ Push data to next stage, ☒ Are initiated by upstream
20. ☒ ClassX, ☒ ClassY, ☒ ClassZ
21. ☒ Final processing unit, ☒ Output storage
22. ☒ Source → Filter1 → Filter2 → Sink, ☒ UI → Business Logic → Data Access
23. ☒ Flexible architecture, ☒ Modular design, ☒ Layer independence
24. ☒ Invoicing B. Logic, ☒ ShoppingBasket B. Logic, ☒ Order B. Logic
25. ☒ Handle timing issues, ☒ Separate filters
26. ☒ Transformed, ☒ Filtered
27. ☒ Push filters, ☒ Source to Filter1, ☒ Filter2 to Sink
28. ☒ Order Data Access, ☒ Payment Data Access, ☒ ShoppingBasket Data Access
29. ☒ Collects data, ☒ Stores final results
30. ☒ Be maintainable, ☒ Reduce redundancy, ☒ Allow easy updates

True or False Questions

1. The architectural design phase defines the overall system structure and behavior.
2. The layered architecture allows each layer to directly access all other layers.
3. The business logic layer is responsible for managing data access and storage.
4. In a pipe-and-filter architecture, filters should modify data in place to save memory.
5. Push-based filters send data forward without being requested.
6. Pull-based filters operate by requesting data from previous filters.
7. A filter can both push and pull data depending on the architecture.
8. The pipe-and-filter model is suitable for applications involving real-time event handling.
9. Buffers can be used in pipes to synchronize data flow between filters.
10. The main advantage of the pipe-and-filter architecture is its support for incremental processing.
11. In a layered architecture, UI logic should be placed in the data access layer.
12. The presentation layer handles the display of information to users.
13. The layered model can be combined with other architectural styles like pipe-and-filter.
14. In the client-server model, clients provide data access services to servers.
15. The data sink is the origin point of data in a data flow system.
16. Filters in a data flow model are typically stateful components.
17. A thread can be used to connect multiple filters in a pipeline.
18. Pull architecture requires the sink to initiate data flow.
19. User interfaces belong in the business logic layer for better separation of concerns.
20. Each layer in a layered architecture should only communicate with adjacent layers.
21. Pie charts, bar charts, and tables are used to represent architectural logic.
22. MVC stands for Model-View-Controller.
23. Models in MVC handle data and business logic.
24. Views in MVC are responsible for user input and data processing.
25. In the architecture diagrams, “UI” refers to User Interface.
26. Data can flow in both directions in a pipe-and-filter model.
27. Components like Customer, Order, and Payment are part of functional domains.
28. Class diagrams show the runtime behavior of the system.
29. The architectural design process impacts maintainability and scalability.
30. An architecture must be flexible to accommodate future system evolution.

Answer Key

1. True
2. False
3. False
4. False
5. True
6. True
7. True
8. False
9. True
10. True
11. False
12. True
13. True
14. False
15. False
16. False
17. True
18. True
19. False
20. True
21. False
22. True
23. True
24. False
25. True
26. False
27. True
28. False
29. True
30. True

Complete the Sentence Questions

1. The _____ of a system defines its overall structure in terms of components and their interactions.
2. A _____ architecture emphasizes separation between presentation, business logic, and data.
3. A layer that manages user input and output is often called the _____ layer.
4. The _____ layer typically handles business rules and processing.
5. The data management responsibilities are handled by the _____ layer.
6. In a push model, data is actively _____ from the source to the sink.
7. In a pull model, data is _____ by the sink from the source.
8. The architectural model using sequential data processing stages is known as a _____ architecture.
9. Filters in a pipe-and-filter architecture can either push or _____ data.
10. Pipes are used to _____ data between filters.
11. The pipe-and-filter model is especially useful for _____ data transformations.
12. The layered architecture promotes separation of _____.
13. The three-layer architecture includes presentation, business logic, and _____ access.
14. In client-server architecture, the server provides _____ to the clients.
15. The architecture where each data transformation step is independent is the _____ model.
16. In a layered system, higher layers use services provided by _____ layers.
17. Data is processed by a filter and then _____ to the next filter or sink.
18. Buffers are often used to handle _____ between filters.
19. The order, customer, and payment systems are examples of _____ domains.
20. User interfaces are typically part of the _____ layer.
21. In the diagrams, "UI" stands for _____ Interface.
22. Business _____ is the layer responsible for decision-making rules.
23. Architectural design is about creating a system's high-level _____.
24. Data _____ is concerned with how data is stored and retrieved.
25. A pipeline architecture is good for parallel _____.
26. In a layered architecture, changes to one layer should not affect other _____.
27. An order, once confirmed, is sent to the _____ system.
28. The MVC pattern separates model, view, and _____.
29. Architectural styles provide reusable _____ for system organization.
30. The main goal of architecture is to support system _____ and evolution.

Answer Key

1. architecture
2. layered
3. presentation
4. business
5. data
6. pushed
7. pulled
8. pipe-and-filter
9. pull
10. transfer
11. streaming
12. concerns
13. data
14. services
15. pipe-and-filter
16. lower
17. forwarded
18. timing
19. functional
20. presentation
21. User
22. logic
23. structure
24. access
25. processing
26. layers
27. invoicing
28. controller
29. patterns
30. maintenance